School/Faculty	Faculty of Computing	Page:	1 of 5
Programme Name	Bachelor of Software Engineering with Honours		
Course code:	SCSR2213	Academic Session/Semester:	20242025/2
Course name:	Network Communications	Pre/co requisite:	NIL
Credit hours:	3		

COURSE INFORMATION

Course synopsis	This course will discuss the basic topics of computer network and data communications. Based on TCP/IP Internet protocol stack, the course will apply top-down approach. Starts with the important and usage of computer network in commonly applications, the approach will go further detail in the technical aspect in data communication. At the end of this course, students will have an understanding and appreciation of how the network works.
Course coordinator (if applicable)	Ms. Hazinah Kutty Mammi

Course lecturer(s)/Section	Name	Office	Contact no.	E-mail @utm.my
02,03	Ms. Hazinah Kutty Mammi	N28-347-06		hazinah

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO	PLO (Code)	Taxonomies and generic skills	T&L methods	Assessment methods
CLO1	Apply the principles of computer networking and the operations of a computer network based on TCP/IP Internet protocol stack.	PLO 1 (KW)	C3	Lecture Active learning	T, F, GR
CLO2	Inspect network using Wireshark and Packet Tracer software and hardware tools.	PLO 2 (CG)	C4	Lab Work	L
CLO3	Work together with different people to design a small local area network based on a real-world case study.	PLO 4 (IPS)	C4, CS7	Group project-based work	GR, PR

T – Test; L – Lab, GR – Group Project; PR – Personal Report; F – Final Exam

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class activities
2.	Lab Work	Conducted using software and simulation tools
3.	Group project-based work	Conducted through case study assignment. Tasks are given in sequential steps throughout the semester. Students in a group of 3 are given a case study that requires computer network knowledge involving the design of a simple local area

Prepared by:		Certified by:
Name:	Ms. Hazinah Kutty Mammi (Course Coordinator)	Name: Prof. Dr. Asri Md Ngadi
Signature:		Signature:
Date:	28 February 2025	Date:

		network and calculations of IPv4 subnetting. The report must comply to the case study specifications and be given in the form of written report.

Weekly Schedule:

Week 1	**Chapter 1: Computer Networks and the Internet** *Group Project Briefing* What is Internet? The Network Edge, The Network Core, Delay, Loss, and Throughput in Packet-Switched Networks
Week 2	**Chapter 1: Computer Networks and the Internet** Protocol Layers and Their Service Models, History of Computer Networking and the Internet
Week 3	**Chapter 2: Application Layer** Principles of Network Applications, The Web and HTTP, Electronic Mail in the Internet, DNS
Week 4	**Chapter 2: Application Layer** *Lab 1* Peer-to-Peer Applications, Video Streaming and Content Distribution Networks
Week 5	**Chapter 3: Transport Layer** *Test 1* Introduction and Transport Layer Services, Multiplexing and Demultiplexing, Principles of Reliable Data Transfer
Week 6	**Chapter 3: Transport Layer** Connection-Oriented Transport: TCP
Week 7	**Chapter 3: Transport Layer** *Lab 2* Principles of Congestion Control
Week 8	**MID-SEMESTER BREAK**
Week 9	**Chapter 4: The Network Layer: Data Plane** Overview, What's Inside a Router? The Internet Protocol (IP): IPv4 Datagram Format
Week 10	**Chapter 4: The Network Layer: Data Plane** *Lab 3, Test 2* IPv4 Datagram Fragmentation, IPv4 Addressing
Week 11	**Chapter 4: The Network Layer: Data Plane** Network Address Translation (NAT), IPv6

Week 12	**Chapter 5: The Link Layer and LANs** Introduction, Error-Detection and Correction Techniques
Week 13	**Chapter 5: The Link Layer and LANs** *Lab 4* Multiple Access Links and Protocols, Switched LAN.
Week 14	**Chapter 6: Wireless and Mobile Networks** *Lab 5* Wireless Links and Network Characteristics, WiFi: 802.11 Wireless LANs
Week 15	**Chapter 6: Wireless and Mobile Networks** Cellular Internet Access, Mobility Management: Principles, Mobile IP, Managing Mobility in Cellular Networks.

Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline	Guided Learning (Face to Face)				Teaching and Learning Activities: Guided Learning Non-Face to Face	Teaching and Learning Activities: Independent Learning Non-Face to face	TOTAL SLT
CLO	**L**	**T**	**P**	**O**			
CLO 1	12		3	3	6	16	**40h**
CLO 2	12	3		3	6	9	**33h**
CLO 3	4		1	1	10	15	**31h**
Total SLT	**28**	**3**	**4**	**7**	**22**	**40**	**104h**

L: Lecture, T: Tutorial, P: Practical, O: Others

Continuous Assessment		PLO	Percentage	Total SLT
1	Lab 1 - 5	PLO2 (CG)	20	10h
2	Test 1	PLO1 (KW)	15	1h30m
3	Test 2	PLO1 (KW)	15	1h30m
4	Group Project and Individual Report	PLO4 (IPS) PLO2 (CG	15	As in CLO3 (5h)
Final Assessment			Percentage	Total SLT
1	Final Examination	PLO1 (KW)	35	3h
Grand Total			**100**	**120h**

*To pass this course, students must pass both Continuous and Final assessment.

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

Packet Tracer Software, WireShark Software, Network Cables & Tester

Learning resources:

Text book (if applicable)

James F. Kurose and Keith W. Ross, Computer Networking: A Top-Down Approach: International Edition (8th Edition), Pearson Education, 2021. ISBN: 9781292405469 (8th); ISBN-13: 978-0133594140 (7th).

Main references

Behrouz A. Forouzan and Firouz Mosharraf, Computer Networks: A Top-Down Approach, McGraw Hill, 2012.

Additional references

Dye, McDonald and Rufi (2011). Network Fundamentals: CCNA Exploration Companion Guide, CISCO Press.

Online

http://elearning.utm.my

Academic honesty and plagiarism: *(Below is just a sample)*

Lab works are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES) Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and disciplinary steps may be taken by the faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

1. Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.
2. Students are required to behave and follow the University's dressing regulation and etiquette all the time.
3. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
4. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted.
5. Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.
6. To pass this course, students must pass both Continuous and Final assessment. Minimum passing marks for Continuous Assessment is 30% while for Final Assessment is 20%. Example: if Continuous Assessment total marks is 65%, students will need a minimum of 19.7% to pass.
7. If the final marks total is less than 40%, the student will get a FAIL grade.

No	Assessment	PLO1 (KW)	PLO2 (GC)	PLO2 (GC)	PLO4 (IPS)	TOTAL
		CLO1	CLO1	CLO2	CLO3	
1	LABS (5)			20		20
2	TEST 1	15				15
3	TEST 2	15				15
4	FINAL EXAM	35				35
5	PROJECT REPORT		5		5	10
6	PERSONAL REPORT				5	5
TOTAL PLO		65	5	20	10	100

Disclaimer:

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